

BILAL SHAKER

Ph.D. | Computational Drug Discovery · AI/ML · Bioinformatics · Biomedical Engineering · Cheminformatics
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PROFESSIONAL SUMMARY

Accomplished computational researcher with 6+ years of experience in machine learning-driven drug discovery, structural bioinformatics, and precision medicine. Developer of globally adopted predictive tools (LightBBB, LogBB_Pred) with 2,000+ users. Author of 25+ peer-reviewed publications with 1,520+ citations and h-index of 14. Alberta Innovates Postdoctoral Fellow with a strong track record of independent research, international collaboration, and mentorship. Committed to building AI-powered platforms that accelerate the translation of computational insights into therapeutic solutions.

CURRENT POSITION

Postdoctoral Researcher – Alberta Innovates Fellow

Sep 2025 – Present

University of Alberta, Edmonton, Canada

- Awarded Alberta Innovates Postdoctoral Recruitment Fellowship — one of Canada's most competitive merit-based research fellowships for early-career scientists
- Developing next-generation AI-powered methodologies for accelerated drug discovery, integrating generative models, transformer architectures, and graph neural networks
- Building and deploying generative AI pipelines for de novo compound generation and lead optimization, extending discovery beyond traditional chemical libraries
- Developing an AI-based drug target identification platform integrating multi-omics data — genomics, transcriptomics, and proteomics — to nominate and prioritize novel therapeutic targets
- Applying diffusion-based generative models and reinforcement learning to design CNS-penetrant compounds optimized for potency, selectivity, and ADMET safety
- Leading collaborative projects bridging computational chemistry and translational research with experimental biochemists and medicinal chemists
- Supervising Master's and PhD candidates in AI/ML methods and computational drug discovery

PROFESSIONAL EXPERIENCE

Research Fellow – Brain Pool Program

Jul 2023 – Aug 2025

Ewha Womans University, Seoul, South Korea

- Awarded the Brain Pool Fellowship by the National Research Foundation of Korea — South Korea's most prestigious national research appointment for international scientists
- Designed and developed machine learning frameworks for rational drug discovery targeting GPCRs and allosteric modulators
- Built deep learning models for protein-ligand interaction prediction and binding affinity estimation, enabling more accurate virtual screening of drug candidates

- Applied molecular docking and molecular dynamics simulations to validate computational predictions and guide experimental prioritization
- Co-authored 8+ peer-reviewed publications in high-impact journals, contributing to advances in computational drug design and structural bioinformatics
- Mentored Master's and PhD students in machine learning methods, drug discovery workflows, and scientific writing
- Transitioned research focus toward generative AI-driven molecular design, establishing the foundation for current work in de novo compound generation

Doctoral Researcher – Bioinformatics

Aug 2019 – Feb 2023

Chung-Ang University, Seoul, South Korea

- Developed LightBBB — a LightGBM-based ML model for blood-brain barrier penetration prediction, published in Bioinformatics (IF 6.931, 214+ citations)
- Developed LogBB_Pred — a quantitative ML model for logBB value prediction, published in Bioinformatics (IF 6.931, 70+ citations)
- Both tools collectively adopted by 2,000+ researchers globally as open-access prediction servers
- Published 15+ peer-reviewed articles with cumulative impact factor exceeding 60
- Conducted comprehensive research across machine learning, drug discovery, protein engineering, and immunoinformatics
- Awarded Best Research Graduate Award for exceptional doctoral contributions
- Recipient of CAYSS Scholar Award — Chung-Ang University Young Scientist Scholarship (2019–2021)
- Supported by Korean Government Research Grant (MSIT/NRF, 2019–2021)

Visiting Lecturer

Sep 2018 – Jul 2019

SARHAD University, Islamabad Campus, Pakistan

- Delivered lectures on Bioinformatics, Parasitology, and Gene Manipulation to undergraduate and postgraduate students
- Collaborated with faculty to promote research-led teaching and application-focused curriculum

EDUCATION

Ph.D. in Bioinformatics

Feb 2023

Chung-Ang University, Seoul, South Korea

Thesis: Development of machine learning-based methods to predict blood-brain barrier permeability | Award: Best Research Graduate Award

M.Phil. in Bioinformatics

Oct 2018

Quaid-I-Azam University, Islamabad, Pakistan

Thesis: Elucidation and screening of drug targets against LpxA of Klebsiella pneumoniae | Award: Punjab Government Scholarship

B.S. in Bioinformatics

Sep 2016

COMSATS Institute of Information Technology, Sahiwal Campus, Pakistan

TECHNICAL SKILLS & EXPERTISE

- Deep learning architectures including CNNs, RNNs, LSTMs, Transformers, and Graph Neural Networks (GNNs) using TensorFlow, PyTorch, and scikit-learn
- Generative AI for molecular design: diffusion models, Variational Autoencoders (VAEs), and reinforcement learning for multi-objective molecular optimization
- AutoML frameworks, multi-task learning, active learning pipelines, and closed-loop computational-experimental discovery systems
- Structure-based drug design (SBDD) and ligand-based drug design (LBDD) for hit identification and lead optimization
- Molecular docking and virtual screening using AutoDock, Vina, and Glide across diverse target classes including GPCRs and allosteric sites
- ADMET property prediction and optimization including BBB permeability, metabolic stability, hERG toxicity, and solubility modeling
- Cheminformatics workflows using RDKit, MOE, and OpenBabel for molecular representation, featurization, and data processing
- De novo compound generation, SAR analysis, pharmacophore modeling, and computer-aided synthesis planning (CASP)
- Protein structure prediction using AlphaFold2, Rosetta, and Modeller for homology and ab initio modeling
- Molecular dynamics simulations using GROMACS and AMBER for conformational sampling, binding pathway analysis, and free energy calculations
- Allosteric site identification, GPCR structural modeling, and protein-ligand interaction analysis
- Multi-omics data integration — genomics, transcriptomics, proteomics — for AI-driven drug target identification and patient stratification
- Protein-protein interaction (PPI) network analysis using graph-based methods for druggability scoring
- Immunoinformatics and multi-epitope vaccine design using reverse vaccinology and MD simulation validation
- Programming: Python (expert), R (proficient), UNIX/Linux, Bash scripting, Git/GitHub, Jupyter Notebooks
- Python stack: NumPy, SciPy, Pandas, Matplotlib, Seaborn, Biopython, Hugging Face Transformers, PyTorch, TensorFlow

PEER-REVIEWED PUBLICATIONS (25+ | 1,520 CITATIONS | H-INDEX 14 | I10-INDEX 18)

2026

1. Yoon, S., Nakliang, P., Shaker, B., Ye, S., Choi, S. Exploring NiCl₂, PdCl₂ and PtCl₂ as a catalyst for [3+2] cycloaddition in Kopsane alkaloid synthesis by density functional theory study. *Medicinal Chemistry Research* 35, 784-791 (2026).

2025

2. Shaker, B., Barakat, K. Harnessing Deep Learning and Generative AI for Molecular Docking Simulations: Tools, Challenges, and Future Directions. *InTechOpen* (2025). [1 citation]

3. Kim, J., Shaker, B., Ko, A., Yoo, S., Na, D., Kang, H.C. Precision medicine approach for in vitro modeling and computational screening of anti-epileptic drugs in pediatric epilepsy patients with SCN2A (R1629L) mutation. *Computers in Biology and Medicine* 191, 110100 (2025). IF=7.0 [4 citations]
4. Han, H., Shaker, B., Lee, J.H., et al. Employing Automated Machine Learning (AutoML) Methods to Facilitate the In Silico ADMET Properties Prediction. *Journal of Chemical Information and Modeling* 65(7), 3215-3225 (2025). IF=5.7 [36 citations]

2024

5. Aftab, N., Masood, F., Ahmad, S., et al. An optimized deep learning approach for blood-brain barrier permeability prediction with ODE integration. *Informatics in Medicine Unlocked* 48, 101526 (2024). [30 citations]
6. Rashid, H., Ullah, A., Ahmad, S., Shaker, B., et al. Identification of novel genes and pathways of Ovarian Cancer using a Comprehensive Bioinformatic Framework. *Applied Biochemistry and Biotechnology* 196(6), 3056-3075 (2024). [8 citations]
7. Umair, M., Ahmed, Z., Shaker, B., et al. A novel homozygous FAM92A gene (CIBAR1) variant further confirms its association with non-syndromic postaxial polydactyly type A9 (PAPA9). *Clinical Genetics* 106(4), 488-493 (2024). [5 citations]

2023

8. Shaker, B., Lee, J., Lee, Y., Yu, M.S., et al. A machine learning-based quantitative model (LogBB_Pred) to predict the blood-brain barrier permeability (logBB value) of drug compounds. *Bioinformatics* 39(10), btad577 (2023). IF=6.931 [70 citations]
9. Ullah, A., Shahid, F.A., Haq, M.U., Shaker, B., et al. An integrative reverse vaccinology, immunoinformatic, docking and simulation approaches towards designing of multi-epitopes based vaccine against monkeypox virus. *Journal of Biomolecular Structure and Dynamics* 41(16), 7821-7834 (2023). IF=2.7 [55 citations]
10. Na, D., Lim, D.H., Hong, J.S., Shaker, B., et al. A multi-layered network model identifies Akt1 as a common modulator of neurodegeneration. *Molecular Systems Biology* 19(12), e11801 (2023). IF=8.5 [12 citations]
11. Kim, J., Teng, L.Y., Na, D., Shaker, B., et al. Genotypes and phenotypes of DNM1 encephalopathy. *Journal of Medical Genetics* 60(11), 1076-1083 (2023). IF=3.5 [7 citations]

2022

12. Shaker, B., Ahmad, S., Shen, J., Kim, H.W., Na, D. Computational Design of a Multi-Epitope Vaccine Against *Porphyromonas gingivalis*. *Frontiers in Immunology* 13, 806825 (2022). IF=8.786 [24 citations]
13. Ud-Din, M., Albutii, A., Ullah, A., Shaker, B., et al. Vaccinomics to Design a Multi-Epitopes Vaccine for *Acinetobacter baumannii*. *International Journal of Environmental Research and Public Health* 19(9), 5568 (2022). IF=4.6 [57 citations]
14. Albekairi, T.H., Alshammari, A., Shaker, B., et al. Design of a Multi-Epitope Vaccine against *Tropheryma whipplei* Using Immunoinformatics and Molecular Dynamics Simulation Techniques. *Vaccines* 10(5), 691 (2022). [33 citations]
15. Alharbi, M., Alshammari, A., Shaker, B., et al. Whole proteome-based therapeutic targets annotation and designing of multi-epitope-based vaccines against *XDR-Alcaligenes faecalis*. *Vaccines* 10(3), 462 (2022). [13 citations]
16. Miandad, K., Ullah, A., Shaker, B., et al. Virtual Screening of *Artemisia annua* Phytochemicals as Potential Inhibitors of SARS-CoV-2 Main Protease Enzyme. *Molecules* 27(22), 8103 (2022). [18 citations]

2021

17. Shaker, B., Ahmad, S., Lee, J., Jung, C., Na, D. In silico methods and tools for drug discovery. *Computers in Biology and Medicine* 137, 104851 (2021). IF=6.698 [760 citations]
18. Shaker, B., Yu, M.S., Song, J.S., Ahn, S., Ryu, J.Y., Oh, K.S., Na, D. LightBBB: computational prediction model of blood-brain-barrier penetration based on LightGBM. *Bioinformatics* 37(8), 1135-1139 (2021). IF=6.931 [214 citations]

19. Shaker, B., Tran, K.M., Jung, C., Na, D. Introduction of advanced methods for structure-based drug discovery. *Current Bioinformatics* 16(3), 351-363 (2021). [17 citations]
20. Khan, S., Shaker, B., Ahmad, S., et al. Towards a novel peptide vaccine for Middle East respiratory syndrome coronavirus and its possible use against pandemic COVID-19. *Journal of Molecular Liquids* 324, 114706 (2021). [13 citations]
21. Thakar, S.B., Ghorpade, P.N., Shaker, B., Lee, J., Na, D. Gas-mediated cancer therapy combined with starvation therapy, ultrasound therapy, chemotherapy, radiotherapy, and photodynamic therapy: a review. *Environmental Chemistry Letters* 19(4), 2981-2993 (2021). IF=13.615 [34 citations]

2020

22. Shaker, B., Yu, M.S., Lee, J., Lee, Y., Jung, C., Na, D. User guide for the discovery of potential drugs via protein structure prediction and ligand docking simulation. *Journal of Microbiology* 58(3), 235-244 (2020). [59 citations]
23. Shaker, B., Ahmad, S., Thai, T.D., Eyun, S., Na, D. Rational Drug Design for *Pseudomonas aeruginosa* PqsA Enzyme: An in silico Guided Study to Block Biofilm Formation. *Frontiers in Molecular Biosciences* 7, 577316 (2020). [33 citations]

2019

24. Ahmad, S., Shaker, B., Ahmad, F., Raza, S.S., Azam, S.S. Molecular dynamics simulation revealed receiver domain of *Acinetobacter baumannii* BfmR enzyme as the hot spot for future antibiotics designing. *Journal of Biomolecular Structure and Dynamics* 37(11), 2897-2912 (2019). [12 citations]

PATENTS

SCN2A (R1629L) mutation epilepsy treatment

Inventors: Na, D., Shaker, B., Ko, A., Kang, H.C., Kim, J.H. | Application No: 10-2024-0112298 | Filed: August 2024

ACTIVE RESEARCH PROJECTS

- De novo drug molecule generation based on protein sequence using deep generative AI models
- Identify novel drug target for gastric intestinal cancer using AI methods
- Allosteric modulator design for GPCRs using rational computational methodology
- Designing novel PROTACs using artificial intelligence: An in silico approach
- Leveraging machine learning for efficient ADMET prediction and generative drug design

AWARDS & HONORS

- Alberta Innovates Postdoctoral Recruitment Fellowship — Alberta, Canada (2025)
- Best Oral Research Presentation — 20th PSK Medicinal Chemistry Workshop, South Korea (June 2025)
- Invited Speaker — ACSGii and ICCST7 International Conferences (December 2023)
- Brain Pool Korea Fellowship — National Research Foundation of Korea (2023)
- Best Research Graduate Award — Chung-Ang University, Seoul, South Korea (2023)
- CAYSS Scholar Award — Chung-Ang University Young Scientist Scholarship Program (2019–2021)
- Research Grant — Korean Government (MSIT/NRF) (2019–2021)
- Punjab Government Scholarship — Quaid-I-Azam University, Pakistan (2016–2018)

- Laptop Award — Prime Minister's Youth Program, Pakistan

PROFESSIONAL REFERENCES

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